HARIKRISHNAN K S

Electronics & Communication Engineering Undergraduate | Analog VLSI | Analog IC Design | FPGA
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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in Analog IC Design and Analog VLSI. Designed a SKY130 CMOS bandgap reference using gm/ID sizing, with DRC/LVS-verified layout in KLayout. Experienced with Xschem, ngspice, KLayout, and LTspice for DC/AC/transient simulation and parasitic extraction. Also skilled in Verilog HDL and FPGA design, seeking entry-level Analog IC Design and Analog Layout roles.

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering *Expected May 2027*
Government Engineering College, Sreekrishnapuram, Palakkad
APJ Abdul Kalam Technological University, Kerala
CGPA : **8.07 / 10** *(up to semester 6)*

TECHNICAL SKILLS

Analog IC Design:	CMOS Analog Design, Semiconductor Devices, gm/ID Methodology, Bandgap Reference (PTAT/CTAT), OTA, Filter Design, Analog Layout, PVT Corners.
PDKs & EDA Tools:	SKY130, GF180MCU, Xschem, ngspice, KLayout, LTspice
Physical Verification:	DRC, LVS, ERC, PEX
Digital Design & FPGA:	Verilog HDL, RTL Design, FSM, Vivado 2022.2, Icarus Verilog, GTKWave, Zynq-7000
Programming:	C, C++, Python
Embedded Systems:	STM32, ESP32, ESP 8266, Arduino, UART, I2C, SPI

TECHNICAL PROJECTS

Design and Layout of CMOS Bandgap Reference Circuit | SKY130 PDK • Xschem • KLayout • ngspice
- Designed and simulated a CMOS bandgap reference using PTAT/CTAT voltage compensation and gm/ID-based transistor sizing, targeting a 1.2 V reference output across −40 °C to 120 °C.
- Implemented a five-transistor OTA, output stage, and startup circuit, targeting startup time below 10 µs and power consumption below 100 µW; performed PVT analysis and verified custom layout using DRC/LVS in KLayout.

Adaptive Real-Time Traffic Light Control Using FPGA | Zedboard • FSM • Verilog • Vivado
- Developed FPGA-based traffic control system in Verilog HDL, improving traffic flow by 25% on Zynq-7000 (ZedBoard) using FSM-based control logic.
- Simulated and verified system behavior using GTKWave and Icarus Verilog.

Enviroguard – Environmental Monitoring System | STM32 • IoT • Proteus
- Designed STM32-based environmental monitoring system, improving power efficiency by 30% with real-time sensor alerts (latency <50 ms).
- Validated circuit design in Proteus prior to hardware implementation.

RFID-Based Attendance System | ESP8266 • IoT
- Developed RFID-based attendance system using ESP8266, enabling 100% contactless tracking and real-time data processing.

WORK EXPERIENCE

Analog IC Design Scholar *Sep 2026 - Present*
CuriousIC Edu (Batch 3)
- gm/ID Design Methodology, GF180MCU PDK, SKY130 PDK, Filter Design, Bandgap Reference Circuits

Analog IC Design Trainee Intern *Jun 2026*
Sophrosyne Semiconductor, Kunjibettu, Udupi
- Trained in analog IC design flow, semiconductor devices, analog building blocks, layout techniques, and physical verification (DRC/LVS/ERC/PEX) Concepts.

CERTIFICATIONS

Advanced Driver Assistance Systems & Machine Learning Workshop *Nov 2023*
Indian Institute of Technology Delhi
Introduction to Internet of Things *Nov 2024*
NPTEL

LEADERSHIP & ACTIVITIES

- Active member, Robotics Club, GEC Sreekrishnapuram; Volunteer, IEEE Young Buddies 2.0; UI/UX Designer & Design Head, ECE Department (incl. department magazine).